Spatial Cluster Randomized Trials - Sampling Design with Spillover Effects & Spatial Dependence

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Abstract

Background: Investigators designing cluster randomized trials desire insights directing treatment assignment methodology for studies ~~in the presence of~~ spillover effects and spatial dependence.

Methods & Simulation: Treatment assignment ~~methods~~ including simple random sampling (SRS) and block stratified sampling ~~(BSS)~~ are defined and spatial autoregressive modeling is applied with consideration for spillover effects and spatial dependence for estimation of intervention effects. A simulation study is carried out comparing SRS and BSS sampling methods on spatial grids of varying sizes. A range of spillover effects and levels of spatial dependence were considered for estimation of the intervention effect via a spatial autoregressive (SAR) model.

Results: Findings of an extensive simulation study comparing simple random sampling and block stratification methods indicate that randomly selected treatment assignments result in reduced average error when estimating intervention effects, but block stratified treatment assignments lead to reduced variation among a series of estimations, indicating that a block stratified treatment arrangement won't achieve the minimal level of estimation error, but it remains robust across a range of selected parameters.

Conclusion: The relationship between spillover effects and mean squared errors (MSE) of intervention effect estimation is apparent. The MSE for the intervention effect is minimized for random sampling treatment assignment, but block stratified assignment minimizes error variance.

1 Introduction

1.1 Background & Motivation

Randomized controlled trials (RCTs) represent the gold standard for establishing causal inference when assessing the efficacy and safety of new interventions[39][41]. However, in many real-world scenarios, particularly in public health, education, and the social sciences, randomizing individuals is not feasible due to logistical constraints, ethical considerations, or the inherent nature of the intervention itself, which may be administered at a community level. In such cases, the cluster randomized trial (CRT) emerges as the preferred and most rigorous design, where intact social units or groups of ~~individuals~~ such as villages, schools, or ~~clinics~~ are randomized to intervention or control arms[23][16].

A key advantage of the CRT design is its ability to evaluate interventions where there is a high risk of contamination between treated and untreated individuals within the same cluster[23][25]. Furthermore, CRTs are uniquely positioned to allow for the estimation of not only the direct effects of an intervention on those who receive it but also the indirect spillover effects. These are the additional impact of an intervention on individuals who did not directly receive it, typically including those in control clusters. Measuring these spillover effects is often of primary scientific and policy interest, as it provides a more ~~complete~~ picture of an intervention's total impact and public health ~~benefit~~[23][22]. When trial clusters are defined by geographic areas, as is frequently the case, the design must contend with inherent spatial complexities that can challenge the core statistical assumption of cluster independence. Two primary and interrelated spatial phenomena can profoundly influence trial outcomes and compromise the validity of the results: spillover effects and spatial dependence[23][41].

Spillover occurs when an intervention's effects in one cluster extend or "spill over"~~,~~ into neighboring clusters, creating significant analytical challenges[2][27]. This is especially common in trials of infectious disease interventions. A study on the use of permethrin-impregnated bed nets by Binka et al. (1998) provided early, compelling evidence of this phenomenon, demonstrating that child mortality was significantly lower in control villages situated near intervention villages, presumably due to a reduction in the local mosquito population[7][20][21]. Similarly, the work by Miguel and Kremer (2004) on a school-based de-worming program in Kenya found significant positive externalities, showing that the health and school attendance of untreated children improved if they lived near schools that were part of the intervention[30]. The critical implication is that failing to account for such positive spillover can lead to a substantial underestimation of the intervention's true effect, as the measured outcomes in the control group are artificially improved by their proximity to the intervention arm. Compounding this issue is spatial dependence, a fundamental concept supported by Tobler's First Law of ~~Geography that indicates~~ all observations are related and nearby observations tend to be more related than distant ones[27][42]. In the context of a CRT, this principle suggests that outcomes in nearby clusters may be correlated due to unmeasured, spatially-patterned environmental, social, or demographic factors. The presence of this spatial autocorrelation violates the crucial assumption that clusters are independent units, which can lead to biased standard errors and ~~inflated~~ Type I error ~~rate which undermines~~ statistical inference[19][15][42].

In response to these challenges, a body of literature has emerged focusing on post-hoc analytical adjustments to account for spatial effects after the data have been collected. A systematic review by Jarvis et al. (2017) found that these existing analytical approaches generally fall into two distinct categories: the spatial variable approach and spatial modeling[27]. The spatial variable approach is the more common of the two. It involves creating a new covariate based on distance or density metrics. Examples include measuring the straight-line distance from control participants to the nearest intervention household (Hawley et al., 2003 & Gimnig et al., 2003)[21][20], calculating outcome rates across different distance bands (Binka et al., 1998)[7], or quantifying the density of treated individuals within a predefined radius (Miguel & Kremer, 2004 & Lenhart et al., 2008)[30][28]. More recently, Chao et al. (2015) developed a "potential exposure" metric to quantify the spatially heterogeneous risk surrounding an individual[14]. In contrast, the spatial modeling approach is more statistically sophisticated, directly incorporating the geographic structure of the data into the random effects component of a statistical method. The method employed by Alexander et al. (2003)[1] considers a distance-decay function in the covariance matrix and an alternative by Silcocks and Kendrick (2010)[40] using spatial weights matrices treats spatial correlation as an underlying unobserved process rather than assuming ~~it can be fully captured by a single, simple metric.~~ Additionally, Watson & Smith (2025) consider an alternative framework from cluster-focused intervention application and instead randomly assign intervention areas and measure "treatment" as a continuous dose which is a function of the observation's distance to the nearest intervention site[45]. Estimation of a dose-response function (DRF) describes how the treatment effect varies over distance from the intervention source rather than directly estimating the intervention or spillover effect in a cluster-based trial.

Despite the growing sophistication of these analytical tools, they are fundamentally post-hoc corrective solutions to a problem that could (and should) be addressed proactively at the design stage. The existing literature reveals a noticeable lack of research into how the initial sampling and allocation of treatments across a geographic landscape can be optimized to manage these known spatial complexities. The review by Jarvis et al. (2017) highlights this gap, noting that in practice, the impact of location on trial results is definitely an afterthought and oftentimes entirely ignored[27]. In fact, trial designs explicitly intended to mitigate spatial effects, such as creating well-separated clusters or buffer zones, were so uncommon that studies employing them were excluded from their primary analysis, underscoring how peripheral these considerations have been to standard practice[27].

This oversight has significant ~~consequences because known~~ studies that ~~do compare spatial analytical approaches~~ consistently find that accounting for spatial effects ~~matters.~~ Silcocks and Kendrick (2010) demonstrated that spatial models not only fit primary care trial data better than ~~the~~ standard CRT models, but also that their inclusion altered both the intervention effect point ~~estimate~~ and its precision[40]. Similarly, Chao et al. (2015) found that adjusting for their spatial exposure metric ~~lowered~~ the estimated effect of a typhoid vaccine[14]. The logical conclusion is that standard trial ~~designs that typically~~ rely on the simple random allocation of clusters, may be systematically producing biased or inefficient estimates without investigators being aware. This leads to the central question motivating the work of our study. That is, "How does the choice of an *a priori* treatment assignment sampling strategy impact the estimation of intervention effects in the presence of both spatial spillover effects and spatial dependence?" While we have methods to analyze the problem after a study is completed, there is currently a lack of evidence-based guidance for investigators on how to design a cluster randomized trial to mitigate these issues from the outset of study planning.

1.2 Aims & Contributions

This study aims to address this critical gap by moving beyond post-hoc analysis to investigate the important role of treatment sampling design in spatial CRTs. Through a comprehensive simulation study, we systematically compare the performance of two contrasting approaches for assigning study clusters to intervention and control arms: simple random sampling (SRS), representing ~~current~~ standard practice for treatment assignment, and a block stratified sampling (BSS) method. The BSS strategy is intentionally designed to create spatial separation between clusters of the same treatment assignment by isolating them from ~~one another to potentially reduce the confounding~~ influence of spillover and spatial dependence.

Our simulations explore how the estimation error of the intervention effect, quantified by Mean Squared ~~Error~~ which is composed of bias and variance, varies across a range of realistic ~~conditions~~ including different magnitudes of spillover effects, the presence or absence of underlying spatial dependence, and variation of restrictions on the geographic distance over which spillover can occur. By identifying the systematic patterns that emerge, this novel research provides practical, evidence-based guidance for investigators designing future spatial CRTs. The findings will equip researchers with the knowledge to select sampling strategies that align with their specific study goals, ultimately leading to more robust, efficient, and valid estimates of intervention effects in complex spatial settings.

The remainder of this article is organized as follows: Section 2 outlines the theoretical framework of the study, including the data generation process, treatment sampling procedures, and estimation methods employed. Section 3 details ~~procedure~~ and results of the simulation study undertaken to evaluate the sampling methods considered. Section 4 considers a ~~real world~~ application of these findings to make ~~a recommendation~~ for other investigators. Finally, Section 5 discusses the implications of our findings, potential drawbacks to the work performed, and potential areas for continued future research.

2 Design of Spatial CRTs

2.1 Cluster Randomized Trials

The randomized controlled trial (RCT) serves as the gold standard design to evaluate intervention efficacy (Kim et al., 2021). Its core principle, the random allocation of individual participants, is expected to balance both measured and unmeasured confounding variables across treatment arms[29]. However, individual randomization is often impractical, or an intervention is delivered at a group level, necessitating an alternative approach[16][23]. In these scenarios, the cluster randomized trial (CRT) is employed, where intact social units or ~~groups like~~ households, clinics, or geographic areas are randomized to intervention arms[16][34]. A key advantage of this design is the prevention of treatment contamination, a scenario ~~where~~ control group members are inadvertently exposed to the intervention, which would otherwise bias the estimated treatment effect toward the null[23]. Statistical considerations in a CRT are the measured response from an intervention and the inherent correlation among outcomes of individuals within the same cluster, a dependency quantified by the intraclass correlation coefficient (ICC)[16][24]. The ICC measures how similar the outcomes of individuals within a cluster are likely to be, relative to those of other clusters. This intra-cluster correlation violates the standard assumption of independence between observations. The failure to account for this feature in the analysis leads to underestimated standard errors and an inflated probability of a Type I error[13][35]. Therefore, the valid design and analysis of CRTs require specialized statistical methods that properly account for this clustering structure, with comprehensive frameworks provided by Donner and Klar (2000)[16] and Hayes and Moulton (2017)[23].

2.2 Neighbor Definitions

2.2.1 Contiguity-Based

Contiguity-based neighbors[17] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other clusters (or cells) that share a common boundary. This boundary may be an edge and/or vertex. The **spdep** R package[8][12][38][10] includes functions for ~~dealing~~ with spatial neighbors and spatial dependence structures. In this study, we consider Rook Neighbors, which fulfill a contiguity condition where a pair of clusters share a common boundary made up by an edge. Another type of contiguity-based ~~neighbors are~~ Queen ~~neighbors that fulfill~~ a contiguity condition where a pair of clusters share a common boundary of either an edge ~~and/or~~ a vertex[17]. Clusters may also have Bishop neighbors where a common boundary at the vertex (corner) of the spatial region is shared. Clusters may also have higher-order neighbors where they do not directly share an edge and/or vertex, but rather have a common boundary with a cluster that is a neighbor of another cluster. In other words, these are “neighbors of neighbors” of the reference cluster.

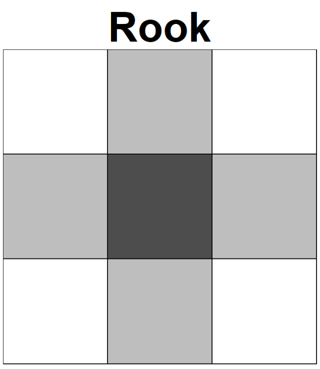

Figure 1: Rook neighbors (light) share an edge with a reference spatial region (dark)[31]

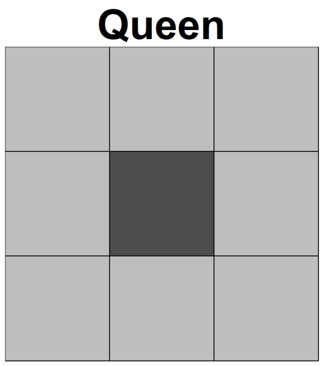

Figure 2: Queen neighbors (light) share either an edge or vertex (or both) with a reference spatial region (dark)[31]

Figure 3: Comparison of Rook and Queen Contiguity Definitions

2.2.2 Distance-Based

Distance-based neighbors[17] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other areas (or cells) that lie within a specified distance of another cell or observation. This distance is commonly ~~delineated~~ by the radius of a circle around the center of a reference spatial region. The region specified in Figure 4 identifies distance-based neighbors of the reference cell (dark) as neighboring cells where the centroid of the cells ~~lie~~ within the circumscribed boundary around the reference cell. In addition to cells, distance-based neighborhoods can also be constructed based on individual points that are within a specified distance of another point.

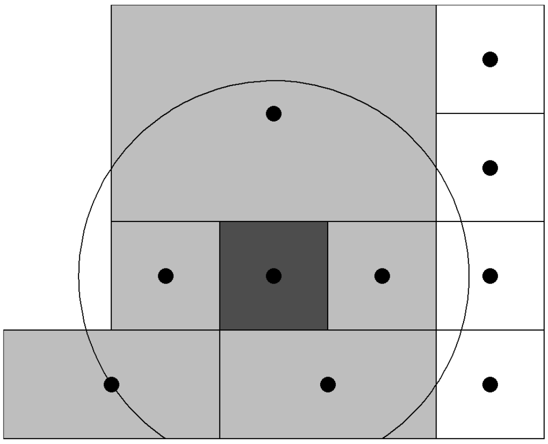

Figure 4: Distance-based neighbors (light) have centroids within a fixed radius of the reference cluster (dark). Clusters with centroids beyond this distance are not considered as neighbors (white)[31]

2.3 Intervention & Spillover Effects

Two primary effect types are considered to understand how treatments affect subjects where they are applied and other subjects in the surrounding areas. The intervention effect and spillover effect are two parameters that are estimated in this study. ~~Spatial cluster randomized trials have subjects or regions that are assigned to receive an intervention and others who are assigned as controls that do not receive an intervention.~~ Observation of the downstream outcomes ~~allow~~ for ~~estimation~~ of the direct intervention effect and indirect spillover effect on the study region.

2.3.1 Intervention Effect

The intervention effect~~(β)~~ represents the estimated effect of the direct intervention applied in a study. This is a measure of the study outcome that is directly attributed to the intervention that is assigned and applied in the study. Estimation of this intervention effect is possible due to the presence of subjects or regions assigned as a control (~~that do not receive an intervention~~) for comparison. ~~Over an array of applications~~ the intervention effect could be a vaccine to prevent infectious disease, a training program for government employees, an educational program for students, and many more. The response y for subject i considering the intervention effect β is expressed as

$$y_i = \alpha + \beta x_i + \varepsilon_i \tag{1}$$

where x_i is an indicator of whether subject i receives the intervention, α is a baseline effect, and ε_i is a random error term for subject i.

2.3.2 Spillover Effect Methods

The spillover effect ~~(ψ)~~ represents an indirect effect observed ~~from the intervention on subjects or regions where an intervention was not directly applied~~[4][5]. There are ~~a number of~~ ways to model a spillover effect as it is observed due to spatial proximity to subjects or regions where an intervention is applied. In this study, only binary spillover effects were considered. A binary spillover effect is a fixed level of a spillover effect that is applied if a subject or region is deemed to be a neighbor of a subject or region where the intervention is applied. ~~The response~~ for subject i considering this ~~spillover effect~~ is expressed as

$$y_i = \alpha + \beta x_i + \psi z_i + \varepsilon_i \cancel{.} \tag{2}$$

where z_i is an indicator of whether the same subject neighbors another subject that receives the intervention ~~(1) or not (0)~~ and is eligible to experience ~~spillover,~~ ~~ψ is~~ the maximum proportion of the intervention effect that can be experienced as spillover - in this case the binary effect applies 100% of possible ~~spillover, and α, β, x_i, & ε_i are as described in Section 2.3.1~~. Alternative methods of modeling a spillover effect include: linear, categorical, exponential, and ~~gaussian~~ decay functions[26][6].

2.3.3 Spillover Effect Types

In this study, there are two specific ways in which spillover of the applied intervention is allowed to propagate to other cells or subjects. In both cases, spillover effects originate from areas where the original intervention treatment is applied. The first case allows for spillover to propagate from cells ~~assigned as receiving~~ the intervention into only cells assigned as control cells. In this case, the effect of the intervention is not permitted to ~~spillover~~ into other intervention cells. This suggests a case where the intervention effect applied is the maximum observable ~~effect~~ and any potential spillover cannot increase the observed effect of a particular intervention. Another method of spillover application is considered where spillover effects are permitted to be applied to both control AND intervention cells that are neighbors of a cell that receives an intervention. This application suggests that an effect greater than that of the original intervention applied can ~~be~~ potentially be observed in a cell that received an intervention and also had spillover from another intervention cell. It is also to be expected that any change in the estimated effects would not be observed in spatial settings where the treatment allocation does not satisfy block stratification conditions (where no intervention cells are rook neighbors with other intervention cells). A representative comparison of these two methods ~~of~~ permissibility of spillover effects is ~~given~~ in Figure 5. In all scenarios of this simulation study, the application of any spillover effect is simply a binary ~~event of~~ whether individuals in a cluster receive a spillover effect or not. Even if spillover exposure could potentially be directed from multiple neighboring clusters, the modeling in this study does not consider any additive spillover effects.

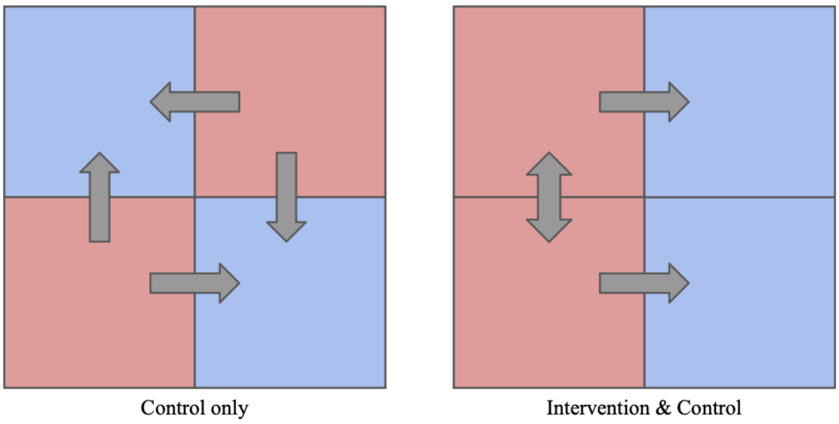

Figure 5: (Left) Spillover application from intervention cells (red) to only control cells (blue) & (Right) spillover application to intervention (red) and control cells (blue).

2.4 Simple Random vs. Block Stratified Sampling

In this study, two different ~~treatments~~ are considered: intervention & control. Assignment of these treatments is considered under two methods: Simple Random Sampling (SRS) & Block Stratified Sampling (BSS). Random sampling treatment assignment considers all possible combinations of intervention & control arrangements on a specified spatial grid and provides equal probability of assignment for any given combination of intervention and control. For example, ~~a 2 row by 3 column spatial grid has 6 total cells.~~ Consideration of a 1:1 ratio between intervention and control treatments ~~provides that~~ three cells ~~are assigned~~ as intervention and three cells are assigned as control. Therefore, there are $\binom{6}{3} = 20$ different combinations of treatments for this specific spatial region.

Block stratified sampling seeks to ensure that for a given balanced number of intervention and control cells, ~~that~~ no intervention or control cells share a common edge boundary ~~with each other~~. An exception to this rule occurs in spatial setups where both dimensions are odd (~~ie.~~ 3x3~~)~~ such that the count of intervention and control cells must be out of ~~balance and we~~ assert that the control class will always be in the majority. In these instances, we relax the common edge boundary restriction for control cells and only require that intervention cells not share a common edge boundary ~~with each other in order~~ to satisfy the block stratification condition.

We now consider a series of spatial grid setups to address efficient estimation of the intervention effect. These spatial grids are ~~denotes~~ as (row $\times$ column) in terms of ~~dimension~~ and we seek to apply a balanced distribution between intervention and control cells where possible. When this isn't possible, we defer to the control group being the majority class ~~(more cells)~~ as mentioned previously.

2.4.1 2x4 Grid

One representative spatial setting considered is ~~grid of 8 individual cells laid out in an arrangement of 2 rows and 4 columns.~~ For this grid setup, a balanced set of intervention and control cells can be assigned where 4 cells are ~~assigned~~ as intervention regions and 4 cells are ~~assigned~~ as control regions. For a grid of 8 total cells and 4 cells of each treatment type, there are $\binom{8}{4} = 70$ unique ~~arrangements that cells~~ can be assigned between treatment and control. Figure 6 (left) displays one of these combinations that satisfies the block stratified condition where intervention and control cells do not have any rook neighbors of the same treatment type. For this spatial setting, there are 2 (out of 70) combinations that satisfy the block stratification condition. Figure 6 (right) displays an example of the 2x4 grid spatial setting with a randomly sampled arrangement of intervention and control cells. Note that both treatment types have cells that are rook neighbors with the same treatment. These spatial grids are constructed ~~with~~ Simple Features for R package[37][38].

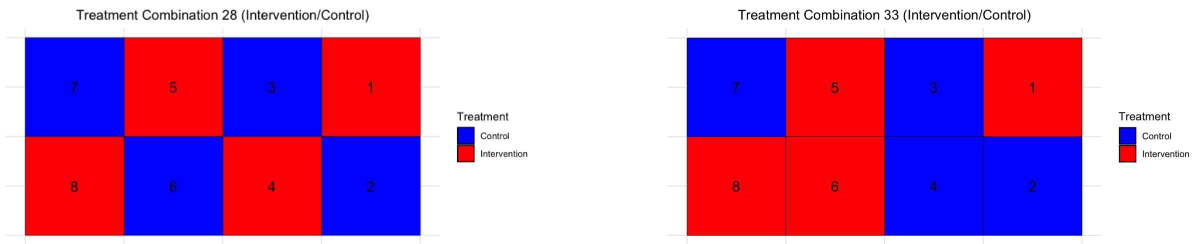

Figure 6: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 2x4 ~~Grid~~

2.4.2 3x3 Grid

The 3x3 symmetric grid setting of 9 cells allows for $\binom{9}{4} = \binom{9}{5} = 126$ unique combinations of intervention and control ~~cells~~ where 5 cells are allocated as control and 4 cells are assigned as ~~treatment~~. The majority class of 5 cells is ~~deemed to be~~ the control for the sake of hypothetical resource management. Figure 7 (left) illustrates one combination of treatment and control cells that completely satisfies the block stratification conditions. Additional cluster combinations satisfy a relaxed condition of block stratification where control cells are allowed to have other control cells as rook neighbors. This is possible in spatial grid settings where both dimensions (row, column) are odd in length. There is one combination for the 3x3 grid setting that satisfies the strongest block stratification condition and 5 more combinations that satisfy the relaxed block stratification condition for a total of 6. An example of ~~randomly~~ sampled treatment assignment that doesn't satisfy block stratification conditions is provided in Figure 7 (right).

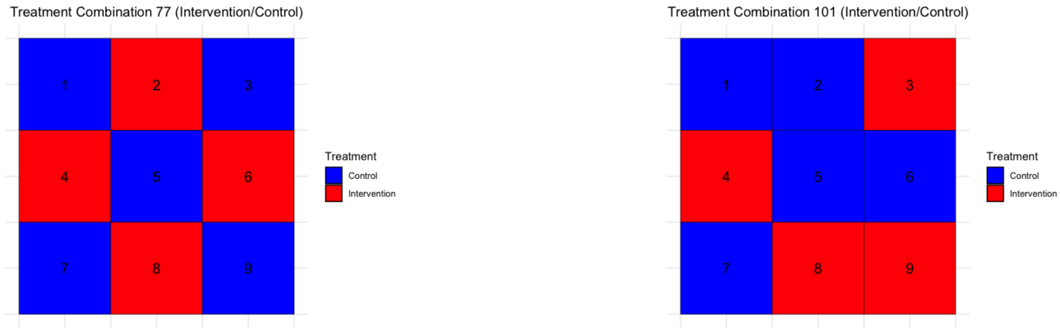

Figure 7: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 3x3 ~~Grid~~

2.4.3 3x4 Grid

The final spatial setting considered was a ~~3 row~~ by ~~4 column~~ (3x4) grid of 12 total cells. This grid, with a balanced allocation of 6 intervention cells and 6 control ~~cells~~ allows for $\binom{12}{6} = 924$ unique arrangements of the two treatments in the spatial region. For this particular grid size, 2 of the 924 treatment combinations satisfy the block stratification ~~condition~~ and Figure 8 (left) presents one of these, with a "checkerboard" design. The remaining treatment combinations are achieved via random sampling of ~~a balanced~~ number of intervention and control cells. Figure 8 (right) illustrates another randomly sampled arrangement of the treatments onto the spatial area where control and intervention cells each have rook neighbors of the same treatment type.

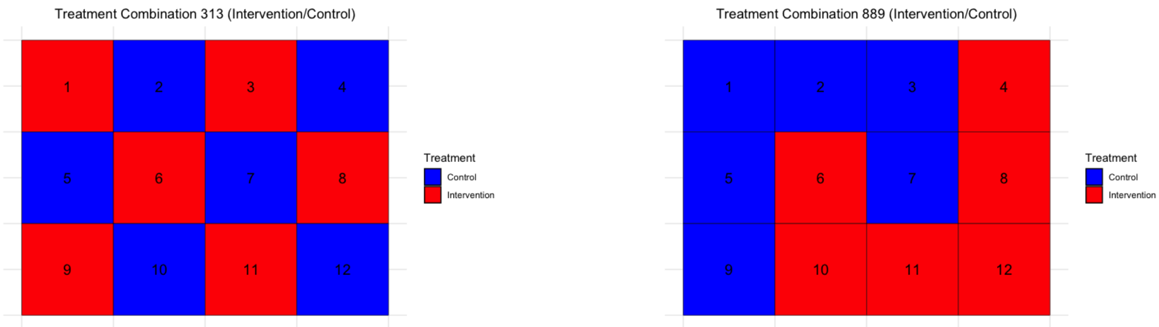

Figure 8: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 3x4 Grid

2.5 Spatial Dependence

Spatial dependence, used interchangeably with spatial autocorrelation, refers to the property of spatial data where observations close to each other tend not be independent of each other. Rather, observations that are close to each other tend to be dependent on each other and influence values of surrounding data. Hence, there is an implied systematic correlation between data at neighboring locations. Spatial correlation can be positive, where similar values tend to be clustered close together, or negative, where dissimilar values are close together. Appropriate consideration of spatial dependence is necessary for performing valid statistical inference as well as for accurate estimation of effects that are modeled. A pair of common methods for assessing spatial autocorrelation are Moran's I [32][33] and Geary's C [18][44]. Moran's I standardizes the spatial auto-covariance by the variance of the data and Geary's C

utilizes the sum of the squared differences between pairs of data as a measure of co-variation. These test statistics are reliant on the spatial weights matrix discussed in Section 2.5.1.

2.5.1 Spatial Weights Matrix

The spatial weights matrix $\mathbf{W}$ is a component of the SAR model from Section 2.6. It is composed elements w_{ij} which indicate the strength or presence of a spatial connection between observations i and j. Therefore the spatial dependence term of the SAR model is represented as

$$\rho \mathbf{W} \mathbf{y} = \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j \tag{3}$$

where ρ is the spatial autocorrelation parameter; an indicator of how strongly neighboring values affect the focal observation, $\mathbf{W}$ is the spatial weights matrix between subjects i and j with matrix elements w_{ij} that are either indicator-based ($\{0,1\}$) or distance-based $\left(\frac{1}{d_{ij}}\right)$ where d_{ij} is the distance between subjects i and j, and $\mathcal{N}(i)$ is the set of all neighbors of observation i. In this study, this set consists of all subjects that are a member of a cluster that is a rook neighbor of the cluster of subject i. For the spatial weights matrix, The indicator-based weights are

$$w_{ij} = \begin{cases} 1, \text{if points } i \text{ and } j \text{ are neighbors} \\ 0, \text{otherwise} \end{cases} \tag{4}$$

and are normalized via

$$W_{ij} = \frac{w_{ij}}{\sum_j w_{ij}} \tag{5}$$

and the distance-based weights are

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}}, \text{if } 0 < d_{ij} < d_{\max} \\ 0, \text{otherwise} \end{cases} \tag{6}$$

which are normalized via

$$W_{ij} = \frac{\frac{1}{d_{ij}}}{\sum_j \frac{1}{d_{ij}}}. \tag{7}$$

2.6 Estimation - Spatial Autoregressive (SAR) Model

Spatial dependence of observations can be considered via modeling with a spatial autoregressive model[44][17][43]. A spatial autoregressive model has the form:

$$y_{ik} = \rho \sum_j w_{ij} y_{jk'} + \mathbf{x}_i \beta + \varepsilon_i \tag{8}$$

where k and k' are different cluster indices for each individual, ρ is the spatial correlation (autoregressive) coefficient, and ε_i represents an i.i.d. error term. This can also be expressed in matrix notation as:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \varepsilon \tag{9}$$

It follows that the general form of the response for this model for the purpose of this study is

$$y_{ik} = \alpha + \beta x_{ik} + \psi z_{ik} + \rho \mathbf{W} \mathbf{y} + \epsilon_i \tag{10}$$

where y_i is the observed response for subject i, α is the baseline/background effect, β is the intervention effect, x_i is an indicator of whether subject i receives the intervention, $\psi(d_i)$ is the spillover effect expressed as a function of distance from the nearest intervention source for subject i, z_i is an indicator of whether subject i can experience any spillover effect, and $\rho \mathbf{W} \mathbf{y}$ captures spatial dependence from neighboring observations and their effects in the response variable. The components of $\rho \mathbf{W} \mathbf{y}$ are ρ, a fixed spatial correlation parameter, $\mathbf{W}$, the spatial weights matrix indicating the strength of relationship between neighboring observations, and the response vector $\mathbf{y}$. Finally, ε_i is a random error term for each subject.

Maximum likelihood estimation is utilized to gather parameter estimates from the spatial lag model[36][11][9]. When considering spatial dependence, the responses are

$$y_i = \rho \sum_{j \neq i} w_{ij} y_j + \alpha + \beta x_i + \psi z_i + \epsilon_i \tag{11}$$

where the residuals are

$$\varepsilon_i = y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i) \,. \tag{12}$$

Then, assuming that $\varepsilon_i \sim N(0, \sigma^2)$, the likelihood function is

$$L\left(\alpha, \beta, \psi, \rho, \sigma^2\right) = \frac{1}{\sqrt{(2\pi\sigma^2)^n \left|\mathbf{I} - \rho\mathbf{W}\right|}} \exp\left(-\frac{1}{2\sigma^2} \epsilon^\top \epsilon\right) \tag{13}$$

where $|\mathbf{I} - \rho\mathbf{W}|$ is the determinant of $(\mathbf{I} - \rho\mathbf{W})$. The log-likelihood is

$$\log L\left(\alpha, \beta, \psi, \rho, \sigma^2\right) = -\frac{n}{2} \log\left(2\pi\sigma^2\right) + \log\left|\mathbf{I} - \rho\mathbf{W}\right| - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i)\right)^2 \tag{14}$$

which is maximized with respect to $\alpha, \beta, \psi, \rho,$ and σ^2. In applied simulation tasks, estimates can be obtained via the 'lagsarlm()' function from the R spdep library[8].

3 Simulation Study

A simulation study was carried out to consider the contrast of block stratified and random treatment assignment sampling methods with consideration for spillover effects, spatial dependence, and restrictions on spillover application between intervention and control regions. The results of this study are detailed in the remainder of this section.

3.1 Simulation Scenarios & Response

We carry out a simulation study to explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence or absence of spatial dependence, varying restrictions on clusters where spillover is permitted to occur, and the method of sampling for treatment assignment. Each simulation scenario is composed of a set of these parameters. The baseline effect, α, is always assigned as 0.2. The intervention effect, β, is always assigned as 1.0. The spillover effect, ψ, has possible values 0.5, 0.6, 0.7, and 0.8. The spatial dependence parameter, ρ is set as either 0.00 or 0.01. The sampling design for each scenario is either Block Stratified Sampling (BSS) or Simple Random Sampling (SRS). Finally, the spillover application type is (1) Control Clusters Only or (2) Control & Intervention Clusters. Each simulation setting is replicated for each of the spatial grid dimensions discussed previously (2x4, 3x3, and 3x4).

Given that the form of the spatial lag model in Section 2.6 contains the response on both sides of the expression, a separate closed-form expression for data generation must be considered[17][3]. This data generation expression is

$$\mathbf{y} = (\mathbf{I} - \rho\mathbf{W})^{-1} (\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \epsilon) \tag{15}$$

where $X\beta = \beta x_i + \psi z_i$. x_i is an indicator of whether subject i receives the intervention and z_i is an indicator of whether subject i receives any spillover effect.

3.2 Simulation Procedure

The algorithm implemented to perform the simulations for this study is illustrated in the following section. The simulation is performed over all potential cluster treatment combinations (intervention/control) for each of the 2x4, 3x3, and 3x4 cluster spatial grid geometries. The simulation procedure is as follows:

1. Define the spatial grid geometry & set of parameters for simulation scenario (α, β, ψ, ρ)
2. List out all cluster treatment combinations for specified spatial grid (70 combinations for 2x4, 126 combinations for 3x3, 924 combinations for 3x4
3. Construct the spatial grid object for the corresponding number of clusters (8, 9, 12) and define the rook neighbors for each cluster in the spatial region.
4. Identify cluster treatment combinations for the specified spatial grid object that satisfy Block Stratified Sampling (BSS) conditions (2x4: 2, 3x3: 6, 3x4: 2).
5. For each combination in the set of all cluster treatment combinations for a given spatial grid:
   (a) Sample 20 subjects per cluster over the entire space of the grid (2x4: 160 subjects, 3x3: 180 subjects, 3x4: 240 subjects).
   (b) Assign each subject to its corresponding cluster and to the intervention or control treatment based on the treatment assignment of the specific cluster.
   (c) Identify the rook neighbors of each subject and whether each subject is exposed to a binary spillover effect based on the treatment assignment of neighboring clusters and selected spillover application type as described in Section 2.3.3.
   (d) Generate the response value for each subject with consideration for spatial dependence via Spatial Autoregressive (SAR) modeling.
   (e) Estimate and retain parameter values for $\alpha, \beta, \psi,$ and ρ with the Spatial simultaneous autoregressive lag model ('lagsarlm()') via maximum likelihood estimation.
   (f) Iterate parts (a)-(e) of Step 5 N=50 times for each cluster treatment combination for the 2x4 and 3x3 spatial grids and N=10 times for each cluster treatment combination for the 3x4 spatial grid.
6. Aggregate parameter estimates for each iteration of each cluster treatment combination for a specified spatial grid
7. For each parameter estimate (α, β, ψ, ρ), compute the bias of the mean of the estimates relative to the true value, variance of the estimates, and Mean Squared Error (MSE) of the estimates such that these metrics are computed for each cluster treatment combination.
8. Repeat Steps 1-5 of the simulation procedure for each spatial grid setup (2x4, 3x3, 3x4).

Figure 9 presents an example of the 8 cluster 2x4 spatial grid object. 160 subjects (20 subjects for each cluster) with uniformly distributed coordinates are sampled over the area of the spatial object and colored according to their cluster membership. Subject counts are not necessarily identical between any pair of clusters as the sample of subjects is considered over the whole space. Points in any cluster on the grid are subject to the selected treatment (intervention/control) and equally experience any potential spillover along with all other points that are members of that same cluster. Similar illustrative figures of the sampled observations over the spatial grid could be constructed and displayed for the other spatial geometries under consideration in the simulation study.

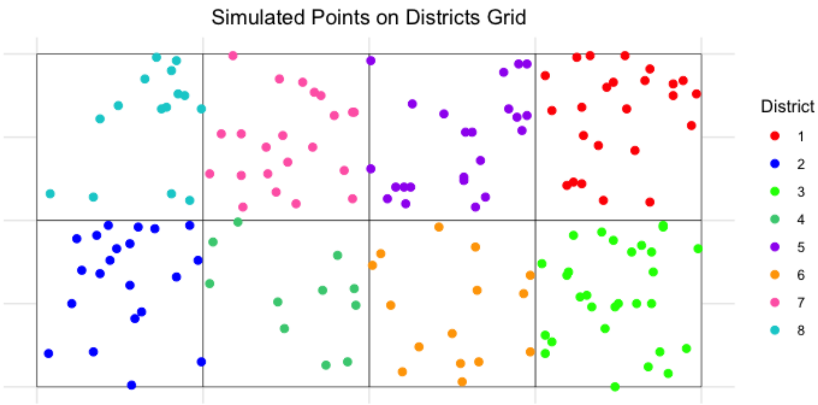

Figure 9: Example of simulated observation coordinates (n=160) on a 2x4 grid of clusters

3.3 Simulation Evaluation Metrics

Estimates for the baseline effect (α), intervention effect (β), spillover effect (ψ), and spatial correlation parameter (ρ) were collected for each iteration of the simulation procedure for every treatment assignment combination. Considering the initial fixed simulation parameters, the following metrics are computed for estimation of the intervention effect for the evaluation of each set of simulation conditions.

- **Bias**: a measure of how different the estimated parameter value is from the true parameter value. Small bias means that the parameter estimate is quite close to the true value.
- **Variance**: a measure of the spread of a set of estimates from multiple simulation iterations. Smaller variance indicates that the estimates are tightly grouped together and large variance means that the estimates are spread out
- **Mean Squared Error (MSE)**: measures the average squared difference between each estimate and the true value of a parameter. Mean Squared Error combines variance to understand the consistency of a set of estimates as well as bias to understand any trend in deviation from the true value into a measure of estimation accuracy.

Mean Squared Error is the primary metric of consideration to understand the efficient estimation of the intervention effect under all simulation settings. The following subsections detail the findings from the simulation study, focusing on the Mean Squared Error (MSE), standard deviation (SD), and bias of the intervention effect (β) estimates under various conditions (simulation scenarios). The analysis compares Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across different spatial grid configurations, spillover effect magnitudes, spatial dependence levels, and spillover application types.

3.4 Intervention Effect Estimates (2x4 grid)

For the 2×4 spatial grid, the simulation results for the intervention effect estimates are presented in Table 1 and visually represented in Figure 10.

When spillover was applied to Control Clusters Only (TrtNoSpill), SRS generally exhibited higher MSE compared to BSS, particularly as the spillover effect (ψ) increased. For instance, with $\rho = 0.00$ (no spatial dependence), SRS MSE ranged from 0.17278 ($\psi = 0.5$) to 0.44075 ($\psi = 0.8$), while BSS maintained very low MSE values (e.g., 0.00790 for $\psi = 0.5$). Similarly, with $\rho = 0.01$ (spatial dependence), SRS MSE increased from 0.30384 ($\psi = 0.5$) to 0.57345 ($\psi = 0.8$), whereas BSS showed MSE values ranging from 0.16515 to 0.56900. BSS consistently demonstrated smaller standard deviations compared to

SRS under this spillover application type, indicating greater precision. SRS estimates generally showed negative bias, which increased in magnitude with higher ψ.

When spillover was applied to Intervention & Control Clusters (TrtSpill), SRS generally showed lower MSE compared to BSS, especially in the absence of spatial dependence ($\rho = 0.00$). For $\rho = 0.00$, SRS MSE ranged from 0.00790 ($\psi = 0.5$) to 0.01872 ($\psi = 0.8$), while BSS MSE was consistently higher (e.g., 0.26160 for $\psi = 0.5$). With spatial dependence ($\rho = 0.01$), SRS still maintained lower MSE (0.09543 to 0.16216) compared to BSS (0.16515 to 0.56900). SRS also exhibited smaller standard deviations and bias magnitudes under this condition, suggesting better overall accuracy and precision.

Across both spillover application types for the 2×4 grid, the presence of spatial dependence ($\rho = 0.01$) consistently led to higher MSE values for both sampling methods compared to scenarios without spatial dependence ($\rho = 0.00$).

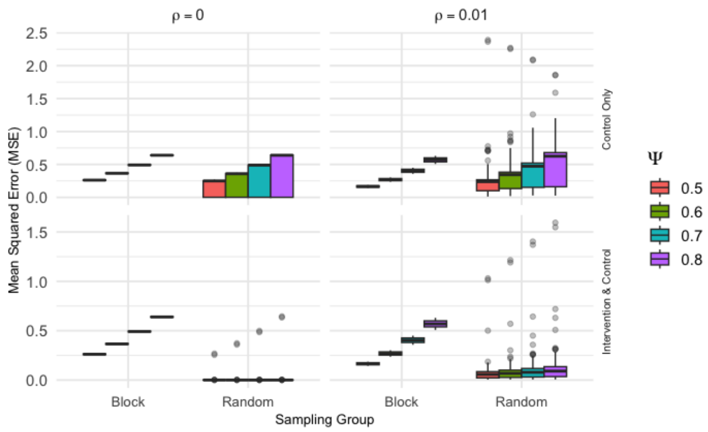

Figure 10: Intervention effect MSE for 2x4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

3.5 Intervention Effect Estimates (3x3 grid)

The results for the 3×3 spatial grid are summarized in Table 2 and Figure 11.

For the Control Clusters Only (TrtNoSpill) spillover application, BSS consistently yielded significantly lower MSE, SD, and bias compared to SRS. For instance, with $\rho = 0.00$, BSS MSE was minimal (e.g., 0.00069 for $\psi = 0.5$), while SRS MSE ranged from 0.09200 to 0.23485. This pattern held true with spatial dependence ($\rho = 0.01$), where BSS MSE remained very low (e.g., 0.00440 for $\psi = 0.5$), contrasting with SRS MSE values ranging from 0.25927 to 0.37154. SRS estimates consistently showed negative bias, increasing with ψ, while BSS bias was negligible.

In the Intervention & Control Clusters (TrtSpill) spillover scenario, BSS continued to demonstrate superior performance with substantially lower MSE, SD, and bias compared to SRS. For $\rho = 0.00$, BSS MSE was consistently around 0.00044, whereas SRS MSE ranged from 0.00233 to 0.00541. With $\rho = 0.01$, BSS MSE was also consistently very low (e.g., 0.03899 for $\psi = 0.5$), while SRS MSE ranged from 0.10339 to 0.16807.

Similar to the 2×4 grid, the presence of spatial dependence ($\rho = 0.01$) generally increased MSE for both sampling methods across both spillover application types for the 3×3 grid.

3.6 Intervention Effect Estimates (3x4 grid)

Table 3 and Figure 12 present the intervention effect estimates for the 3×4 spatial grid.

Under the Control Clusters Only (TrtNoSpill) spillover condition, BSS consistently outperformed SRS in terms of MSE, SD, and bias. With $\rho = 0.00$, BSS MSE was extremely low (e.g., 0.00036 for $\psi = 0.5$), while SRS MSE ranged from 0.03425 to 0.21912. This trend persisted with $\rho = 0.01$, where BSS MSE

Spillover Application	**Spatial Dependence (ρ)**	**Spillover Effect (ψ)**	**Simple Random Sampling**			**Block Stratified Sampling**		
			MSE	SD ($\times 10$)	Bias	MSE	SD ($\times 10$)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.304	4.053	-0.270	0.165	0.294	-0.349
		0.6	0.382	4.062	-0.342	0.269	0.462	-0.471
		0.7	0.471	4.143	-0.414	0.404	0.653	-0.594
		0.8	0.573	4.390	-0.486	0.569	0.867	-0.716
	No (0.00)	0.5	0.173	1.170	-0.343	0.262	0.095	-0.496
		0.6	0.248	1.683	-0.411	0.366	0.100	-0.597
		0.7	0.338	2.291	-0.480	0.492	0.095	-0.697
		0.8	0.441	2.994	-0.548	0.639	0.081	-0.798
Intervention & Control Spillover	Yes (0.01)	0.5	0.095	1.762	0.031	0.165	0.294	-0.349
		0.6	0.115	2.082	0.030	0.269	0.462	-0.471
		0.7	0.138	2.423	0.029	0.404	0.653	-0.594
		0.8	0.162	2.787	0.027	0.569	0.867	-0.716
	No (0.00)	0.5	0.008	0.438	-0.014	0.262	0.095	-0.496
		0.6	0.011	0.614	-0.017	0.366	0.100	-0.597
		0.7	0.015	0.825	-0.020	0.492	0.095	-0.697
		0.8	0.019	1.072	-0.023	0.639	0.081	-0.798

Table 1: SRS vs. BSS: Intervention Effect Error Estimates - 2x4 Spatial Grid Setting

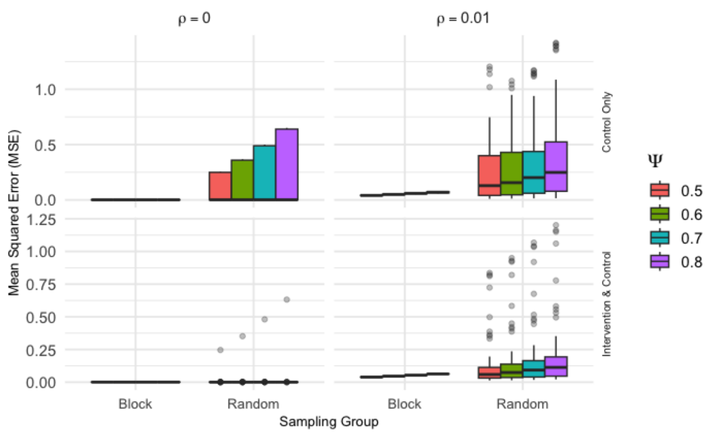

Figure 11: Intervention effect MSE for 3x3 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

remained low (e.g., 0.03463 for $\psi = 0.5$), compared to SRS MSE ranging from 0.12009 to 0.20387. SRS estimates generally showed a small positive or negative bias, while BSS bias was consistently negligible.

For the Intervention & Control Clusters (TrtSpill) spillover application, BSS again showed superior performance with lower MSE, SD, and bias. With $\rho = 0.00$, BSS MSE was consistently around 0.00028, while SRS MSE ranged from 0.00073 to 0.02853. With $\rho = 0.01$, BSS MSE was also low (e.g., 0.22286 for $\psi = 0.5$), while SRS MSE ranged from 0.11193 to 0.19046.

Across all grid sizes and conditions, increasing the spillover effect magnitude (ψ) generally led to an increase in MSE for both sampling methods. The presence of spatial dependence ($\rho = 0.01$) consistently

Spillover Application	**Spatial Dependence (ρ)**	**Spillover Effect (ψ)**	**Simple Random Sampling**			**Block Stratified Sampling**		
			MSE	SD ($\times$10)	Bias	MSE	SD ($\times$10)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.234	2.593	-0.054	0.041	0.044	-0.012
		0.6	0.276	2.860	-0.092	0.050	0.080	-0.018
		0.7	0.322	3.284	-0.131	0.059	0.114	-0.022
		0.8	0.372	3.857	-0.170	0.069	0.145	-0.025
	No (0.00)	0.5	0.092	1.207	-0.182	0.001	0.002	0.003
		0.6	0.132	1.740	-0.219	0.001	0.002	0.003
		0.7	0.180	2.370	-0.255	0.001	0.002	0.003
		0.8	0.235	3.099	-0.292	0.001	0.002	0.003
Intervention & Control Spillover	Yes (0.01)	0.5	0.103	1.480	0.071	0.039	0.022	-0.019
		0.6	0.122	1.688	0.082	0.046	0.027	-0.020
		0.7	0.143	1.905	0.092	0.054	0.032	-0.021
		0.8	0.168	2.149	0.101	0.063	0.007	-0.019
	No (0.00)	0.5	0.002	0.219	-0.004	0.001	0.004	0.004
		0.6	0.003	0.313	-0.004	0.001	0.004	0.004
		0.7	0.004	0.428	-0.005	0.001	0.004	0.004
		0.8	0.005	0.562	-0.006	0.001	0.004	0.004

Table 2: SRS vs. BSS: Intervention Effect Error Estimates - 3x3 Spatial Grid Setting

resulted in higher MSE for both SRS and BSS compared to scenarios without spatial dependence ($\rho = 0.00$). While SRS sometimes showed lower MSE when spillover was applied to both intervention and control clusters, BSS consistently demonstrated lower variance and often lower bias, particularly when spillover was restricted to control clusters only. This highlights that BSS provides more consistent estimates, even if SRS might occasionally yield a lower average error depending on the specific spillover dynamics.

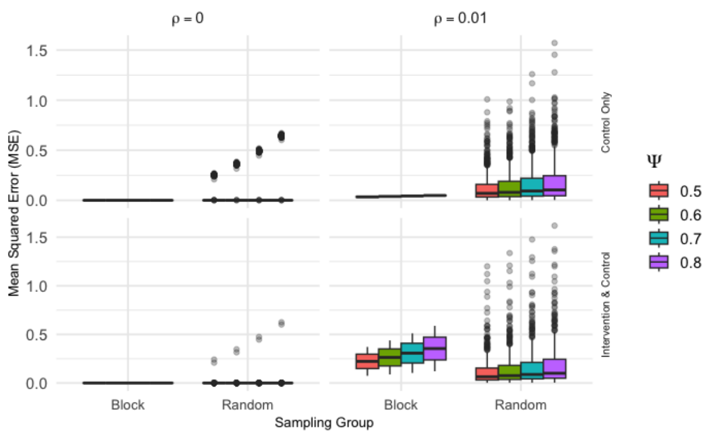

Figure 12: Intervention effect MSE for 3x4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD ($\times$100)	Bias	MSE ($\times$10)	SD ($\times$100)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.120	13.244	0.033	0.346	0.963	-0.152
		0.6	0.138	14.807	0.021	0.388	0.505	-0.167
		0.7	0.159	17.166	0.009	0.435	0.011	-0.180
		0.8	0.183	20.387	-0.001	0.484	0.561	-0.193
	No (0.00)	0.5	0.034	8.554	-0.068	0.004	0.022	-0.006
		0.6	0.049	12.318	-0.081	0.004	0.022	-0.006
		0.7	0.067	16.769	-0.094	0.004	0.020	-0.006
		0.8	0.087	21.912	-0.108	0.003	0.016	-0.006
Intervention & Control Spillover	Yes (0.01)	0.5	0.112	12.873	0.022	2.229	20.995	-0.415
		0.6	0.131	14.762	0.021	2.634	24.750	-0.452
		0.7	0.151	16.817	0.020	3.074	28.774	-0.488
		0.8	0.173	19.046	0.018	3.548	33.060	-0.525
	No (0.00)	0.5	0.001	1.051	-0.001	0.003	0.004	-0.003
		0.6	0.001	1.542	-0.001	0.003	0.004	-0.002
		0.7	0.001	2.143	-0.002	0.003	0.005	-0.002
		0.8	0.002	2.853	-0.002	0.003	0.005	-0.002

Table 3: SRS vs. BSS: Intervention Effect Error Estimates - 3x4 Spatial Grid Setting

4 Application

The findings from our simulation study ~~offer~~ relevant insights to ~~provide guidance for~~ investigators designing CRTs in real-world applications where spatial dependence and spillover effects ~~are potentially~~ present and ~~to~~ be accounted for. ~~In this case, a pair of studies in development that we consider involve the North Carolina Department of Adult Corrections.~~ The structure of correctional systems, particularly community supervision, often involves geographically clustered units where staff from different clusters may interact and where regional similarities can influence outcomes. The investigators leading these studies ~~desire~~ recommendations for treatment assignment ~~methodology~~ ~~regarding~~ the spatially clustered nature of their design.

4.1 Background of Probation in North Carolina

The North Carolina Department of Adult Corrections Division of Community Supervision (NC DCS) provides a clear example of a spatially structured organization, making it an ideal context for applying our research. The DCS is organized hierarchically into four geographic Divisions (East to West), which are subdivided into 30 Districts that make up the entirety of the state. Each district is composed of one or more contiguous counties. In these districts, Probation and Parole Officers (PPOs) supervise individuals at the ~~county level and are~~ themselves managed by Chief Probation/Parole Officers (CPPOs) and Judicial District Managers (JDMs) who oversee districts. Each division (composed of multiple districts) is managed by a Judicial Division Director (JD Dir) as well. A map of the segmentation of each division and district is presented in Figure 13.

This geographic and administrative structure means that PPOs in adjacent counties or within the same ~~district~~ are likely to interact, creating potential channels for spillover effects if an intervention is provided to one group and not another. Furthermore, adjacent counties within the same region

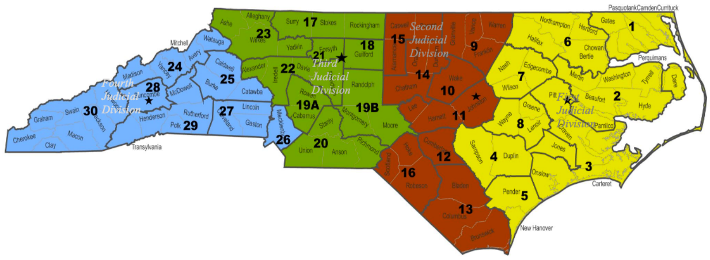

Figure 13: The North Carolina Department of Adult Corrections Division of Community Supervision map is segmented into 30 judicial districts that are grouped into 4 judicial divisions.

often share demographic characteristics, labor market conditions, and similar judicial cultures, leading to spatial dependence in probation outcomes. The choice of treatment assignment strategy in such a context is therefore non-trivial when considering options of either simple random sampling or block-stratified sampling.

4.2 Case 1: Trauma-Informed and Engaged Supervision

~~One~~ proposed study involves implementing a trauma-informed supervision model for PPOs managing specialty caseloads like domestic violence, sex offender, or gang affiliated cases. The intervention consists of intensive training and follow-up sessions delivered to cohorts of PPOs and CPPOs. The primary outcomes would be changes in officer knowledge and attitudes, as well as impacts on individuals under their supervision.

The key design decision for this study is the unit of randomization. Randomizing at the PPO level is not feasible due to the high likelihood of spillover within a county office. Therefore, a cluster randomized design at the county or district level is necessary. ~~Original plans~~ for a similar study randomized at the district level specifically to minimize spillover, acknowledging the risk of contamination between adjacent units. This ~~scenario~~ directly reflects the conditions ~~modeled~~ in our simulation, where the effectiveness of an intervention in one cluster may influence outcomes in neighboring control clusters.

4.3 Case 2: Testing an implementation strategy for specialty mental health probation

~~A~~ second proposed study aims to test a packaged strategy for improving the implementation of Specialty Mental Health Probation (SMHP). The ~~intervention~~ involves enhancing collaboration between PPOs and community-based behavioral health service providers through cross-training sessions and joint treatment team meetings. The primary outcome is improved collaboration, with secondary outcomes related to treatment engagement for individuals on probation.

This case presents a more complex spatial challenge. In addition to the geographic proximity of probation districts, the behavioral health system in North Carolina is managed by four large Managed Care Organizations (MCOs), each responsible for a different catchment area. Many service provider agencies are large, operating in multiple counties and potentially across different MCO regions. This creates multiple, overlapping layers of potential spillover. Figure 14 presents a map of which counties in the state that each MCO operates. ~~An intervention implemented in one county could be adopted by a provider and informally carried into a neighboring control county,~~ or an MCO could promote a similar initiative, ~~confounding the study's results.~~ This complex spatial linkage underscores the importance of a robust treatment assignment strategy.

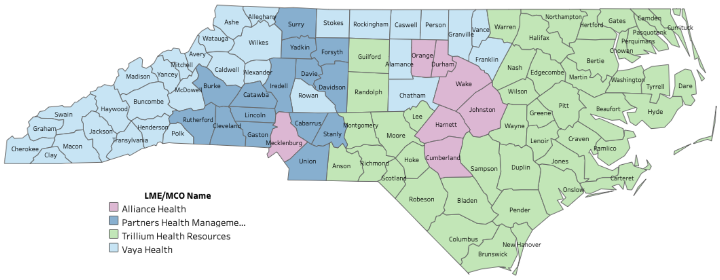

Figure 14: Map of Local Management Entity/Managed Care Organizations (LME/MCOs) for each county in the state of North Carolina. ~~There are 4 LME/MCOs of which one operates in each of the 100 counties of the state.~~

4.4 Recommendations for treatment assignment

The results of our simulation study indicate a clear trade-off: we previously concluded that Simple Random Sampling (SRS) reduces the average mean squared error (MSE) across many randomizations, while Block Stratified Sampling (BSS) reduces the variance of the error. Put simply, BSS is less likely to produce a "perfect" assignment by chance but is also robust against a "worst-case" assignment that could render the study results invalid. For both case studies presented, Block Stratified Sampling (BSS) is the recommended approach.

In Case 1 (Trauma-Informed Supervision), a simple random assignment of districts could, by chance, result in a geographic imbalance, concentrating treated districts in one part of the state. This would confound the intervention effect with regional characteristics. A more robust design would be to stratify by the four state Divisions and randomize districts to treatment and control conditions within each Division. This ensures geographic balance and leverages the finding that BSS minimizes estimation variance, providing a more stable and defensible estimate of the intervention effect.

In Case 2 (SMHP Implementation), the rationale for BSS is even stronger due to the complex potential for spillover. A simple random assignment of counties would ignore the significant structural influence of the MCOs. The most effective design would be to stratify by MCO catchment area, randomizing counties to the implementation strategy or control condition within each MCO. This approach directly controls for the significant confounding variable of MCO-level policies and provider networks. Given the high risk of contamination, the robustness of BSS in minimizing error variance makes it the most prudent choice for achieving a reliable ~~effect estimate~~. ~~While SRS might achieve a lower MSE~~ (intervention error estimate) ~~in theory~~, the risk of a single, poorly distributed randomization in a complex, real-world setting outweighs the potential benefit.

5 Discussion

5.1 Relevance of results

This simulation study provides insights into the efficiency of treatment assignment methodologies in spatial cluster randomized trials (CRTs) when considering spillover effects and spatial dependence. A primary contribution of this work is the direct comparison between Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across various scenarios, offering practical guidance for investigators designing such trials.

Our findings show a clear relationship between the magnitude of spillover effects and the estimation error of the intervention effect. As the spillover effect (ψ) increases, the Mean Squared Error (MSE) for the intervention effect (β) generally rises, regardless of the sampling method or the presence of spatial dependence. This highlights the importance of accounting for spillover effects, as their increasing influence can diminish the accuracy of intervention effect estimates. The results indicate that randomly selected treatment assignments lead to reduced average error (lower bias) when estimating intervention effects, while block stratified treatment assignments consistently result in reduced variation (lower variance) among a series of estimates. This suggests a trade-off: SRS provides a more accurate central estimate on average, whereas BSS offers greater precision and consistency across repeated trials, potentially with higher bias.

The study also clarifies the impact of spatial dependence (ρ) on intervention effect estimation error. The presence of even a small degree of spatial dependence (e.g., $\rho = 0.01$) generally increased MSE for both sampling methods. This indicates that spatial correlation among observations can impede the accurate estimation of intervention effects if not managed during the design phase. The observed patterns across different spatial grid dimensions (2×4, 3×3, and 3×4) support these conclusions, demonstrating the consistency of the findings across varying spatial configurations.

The investigation into cluster spillover restrictions (spillover to control clusters only vs. spillover to both control and intervention clusters) also identified important differences. When spillover was restricted to control clusters only, the MSE for the intervention effect was higher for SRS compared to BSS, particularly with increasing spillover magnitude. Conversely, when spillover was permitted for both control and intervention clusters, SRS often exhibited lower MSE, especially when spatial dependence was absent. This understanding of spillover application types provides investigators with a more precise basis for selecting a sampling strategy that aligns with their specific hypotheses regarding how an intervention's effects might propagate.

In summary, this study offers a comparison of two opposing sampling methodologies in spatial CRTs. The patterns observed regarding spillover magnitude, spatial dependence, and spillover application types provide practical information for investigators. They must consider the goal of an unbiased average estimate (favored by SRS) versus the need for consistent, low-variance estimates (achieved with BSS), especially when designing studies where spillover effects are expected.

5.2 Limitations

While this simulation study provides valuable insights, it is important to acknowledge several limitations that may affect the generalizability and scope of its findings. The study encountered issues related to multicollinearity between the direct (intervention) and indirect (spillover) effects. In some spatial configurations and with specific spillover magnitudes, the high correlation between the xi (intervention indicator) and z_i (spillover indicator) variables in the spatial autoregressive model (Equation 8) could lead to instability in parameter estimates. While the SAR model addresses spatial relationships, extreme collinearity can still impact the precision of individual coefficient estimates, potentially affecting MSE calculations. This issue in separating highly correlated effects is known in spatial causal inference and requires careful consideration in applied settings. It is also necessary to note that the expanded simulation space quickly becomes computationally infeasible. The current study explored a specific range of spillover effect sizes, spatial dependence parameters, and grid dimensions. However, a more extensive exploration of additional parameters (e.g., different baseline effects, varying intervention effect sizes, or a wider range of spatial dependence values) or more complex spatial structures would require significantly greater computational resources. This limitation restricted the number of scenarios that could be thoroughly investigated within the current framework.

An additional notable limitation is the rudimentary spillover mechanism (binary effect) applied. This simulation study modeled spillover as a fixed, binary event, meaning a unit either experienced the full potential spillover effect or none, based solely on contiguity (Rook neighbors). This approach does not account for more realistic scenarios where spillover effects may decay with distance, or where cumulative exposure from multiple treated neighbors could lead to additive or non-linear effects. The absence of more sophisticated distance-based decay functions (e.g., linear, categorical, exponential, or Gaussian decay) or mechanisms for additive spillover limits the direct applicability of these specific quantitative results to real-world situations where such nuanced spillover dynamics are likely. The current model also did not consider higher-order neighbors beyond immediate (Rook) contiguity. The study also utilized homogeneous cluster sizes and regular grid shapes. Real-world clusters often exhibit irregular shapes and varying sizes, which could influence spillover dynamics and the efficiency of different sampling strategies. The current findings may not fully extend to such heterogeneous spatial arrangements. Finally, the study assumes a fixed treatment allocation ratio (primarily 1:1 or near 1:1). Varying the proportion of intervention to control clusters could impact the statistical power and bias of intervention effect estimates, especially in the presence of spillovers.

5.3 Future Considerations

Building on this study's findings, several areas for future research can further improve our understanding of sampling design in spatial cluster randomized trials. A key next step involves incorporating additional spillover structures beyond the binary mechanism. Future simulations should explore various distance-based decay functions, such as linear, categorical/step-wise, exponential, or Gaussian decay. This would provide a more realistic representation of how intervention effects attenuate over space and allow for the identification of optimal sampling strategies under different decay profiles. For example, an intervention with rapid exponential decay might benefit from a different sampling approach than one with a more gradual linear decay.

Furthermore, the current model's assumption of non-additive spillover effects could be relaxed. Future research should consider the implications of "additive" spillover, where being a neighbor to multiple intervention cells allows for a greater cumulative spillover effect. This would reflect scenarios where the intensity of exposure to the intervention's indirect effects scales with the number or proximity of treated neighbors. Understanding how sampling methods perform under such additive spillover mechanisms is important for interventions where collective exposure drives indirect effects. Exploring other neighbor structures beyond Rook contiguity is also warranted. Investigating Queen neighbors (sharing an edge or vertex) or higher-order neighbors (neighbors of neighbors) would provide a more complete understanding of how different definitions of "proximity" influence spillover propagation and, consequently, the performance of various sampling designs.

Additionally, future work could investigate the impact of "prior" weighting of intervention based on cluster subject density. In real-world settings, clusters rarely have uniform population density. Incorporating variable subject densities within clusters and exploring how this influences spillover effects and optimal sampling strategies would add realism to the simulations. An extension of the simulation parameters to cover a broader range of values for spillover effects, spatial dependence, and grid dimensions would allow for a more comprehensive mapping of the parameter space and a more robust set of recommendations for practitioners. This would enable a deeper understanding of the conditions under which each sampling method (SRS vs. BSS) is most advantageous, leading to more efficient and reliable designs for spatial cluster randomized trials.

Future research could also investigate the robustness of findings to model misspecification. This includes evaluating how different sampling strategies perform when alternative spatial models (e.g., Spatial Error Model, Spatial Durbin Model) are used for analysis, or when the true underlying spatial process deviates from the assumed SAR model. Finally, exploring optimal allocation ratios of intervention to control clusters in the presence of spillover effects and spatial dependence could provide valuable insights for maximizing statistical power or minimizing estimation error. Incorporating cost-effectiveness analysis into the design framework would also be beneficial, as different sampling strategies may have varying logistical and financial implications in real-world implementation.

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